Master of Computer Applications

Semester – III

Data Analytics for Business

Final Project

Title: Spotify Music Dashboard

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Spotify Music Dashboard

Developing the Spotify Music Dashboard

GithubLink:

https://github.com/GANESHA08/Data_Analytics_projects/tree/main/Final_Project

Powerbi Service Link:

View BMW Sales Dashboard in Power BI Service

Problem Statement and Data Collection:

Given Problem Statement:

You are a Business Intelligence Consultant for a growing organization. Your task is to develop an end-to-end Business Analytics solution using a dataset of your choice to support data-driven business decisions.

Project-Specific Problem Statement:

The main objective of this project is to develop a Business Intelligence solution for BMW car sales analysis using 50,000 sales records. The project aims to understand sales trends, regional performance, customer preferences, pricing patterns, and high-performing BMW models. MySQL and SQL are used to store and analyze the data, while Power BI is used for data transformation, DAX-based KPI calculations, and interactive dashboard development. The overall goal is to turn raw sales data into clear and useful insights that can support better business decisions.

Data Description:

The dataset used for this project is the BMW Car Sales Classification dataset (BMW_Car_Sales_Classification.csv), which was imported into MySQL and later connected to Power BI for analysis. The dataset contains 50,000 BMW sales records, with each row representing a vehicle sales record.

The main fields in the dataset include:

- **Model** – identifies the BMW vehicle model.
- **Year** – indicates the vehicle model year.
- **Region** – shows the geographical sales region.
- **Color** – represents the vehicle's exterior color.
- **Fuel Type** – specifies the type of fuel used by the vehicle.
- **Transmission** – identifies the transmission type, such as automatic or manual.
- **Engine Size (L)** – indicates the engine capacity in liters.
- **Mileage (KM)** – records the vehicle mileage.
- **Price (USD)** – represents the vehicle price in US dollars.
- **Sales Volume** – shows the number of vehicles sold.
- **Sales Classification** – categorizes the sales performance into different levels.

Business Problem:

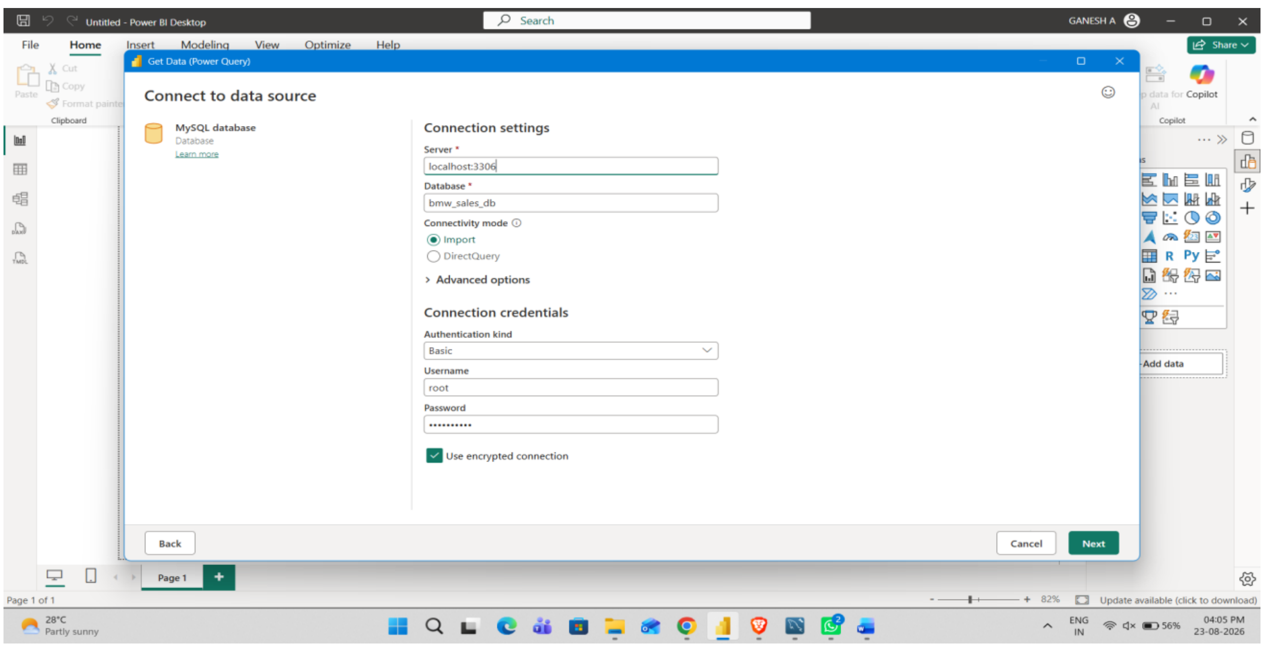
BMW generates large amounts of vehicle sales data across different models, regions, fuel types, transmission types, prices, and vehicle specifications. However, without an integrated analytics solution, it can be difficult to identify sales trends, compare regional and model performance, understand customer preferences, and determine the factors associated with high or low sales.The business problem is to transform BMW sales data into a clear and interactive analytics solution that helps management monitor sales performance, identify high-performing models and regions, analyse pricing and vehicle characteristics, and support data-driven business decisions.

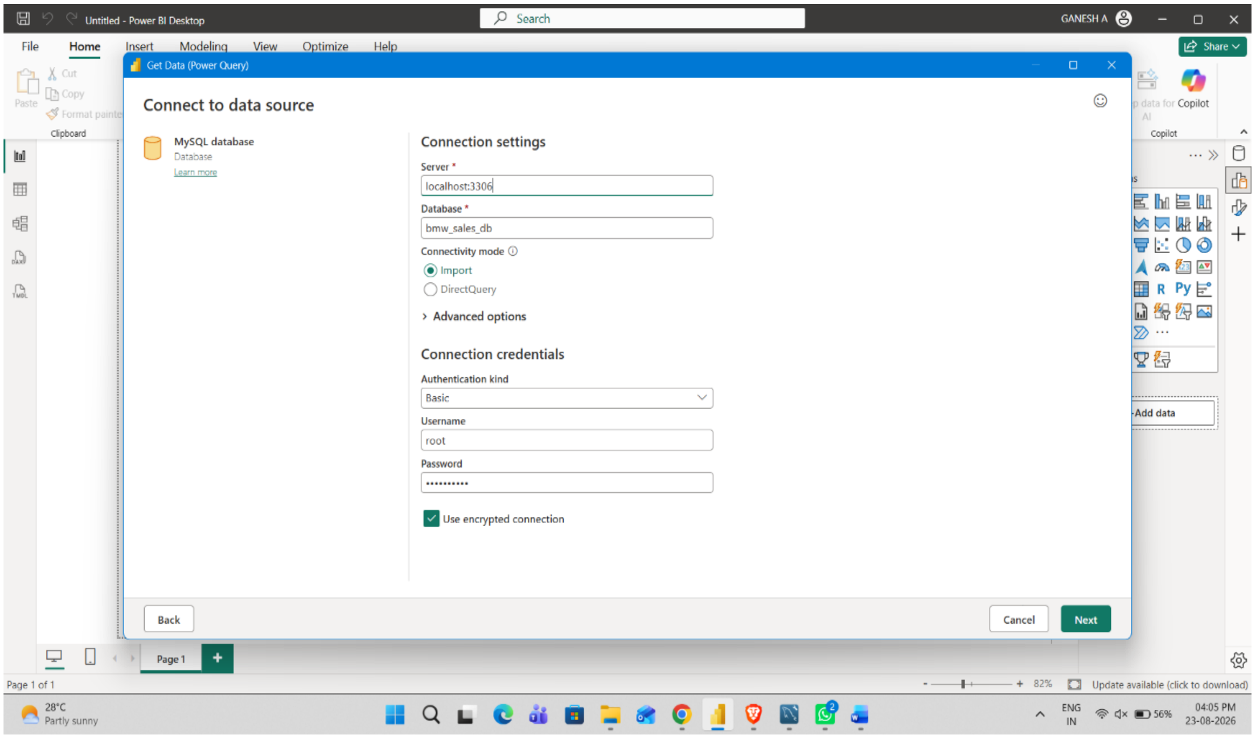
Data Cleaning & Transformation (Power Query) :

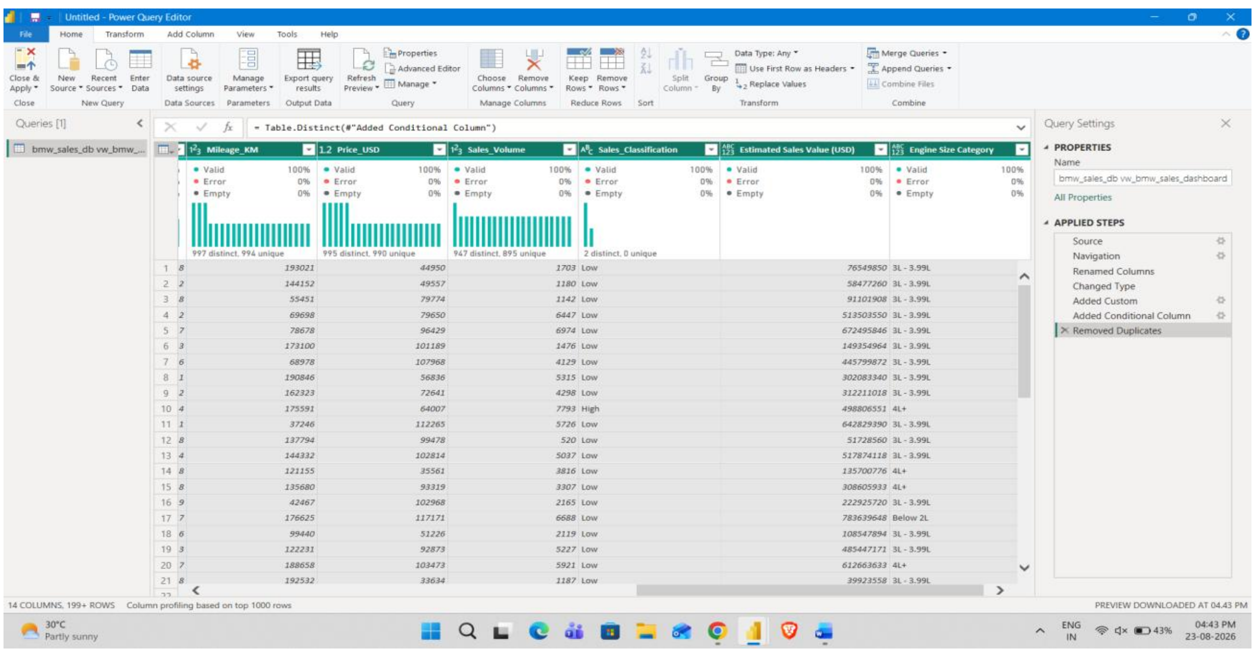
The BMW car sales dataset was connected to Power BI through Get Data → MySQL Database, and the following transformations were performed in Power Query Editor before loading the data into the model:

- Checked the imported data and verified that the BMW sales fields were loaded correctly.
- Verified and assigned appropriate data types to the columns.
- Set Model Year, Sales ID, Mileage (KM), and Sales Volume as Whole Number.
- Set Engine Size (L) and Price (USD) as Decimal Number.

- Kept categorical fields such as BMW Model, Sales Region, Vehicle Color, Fuel Type, Transmission Type, and Sales Classification as Text.
- Created Estimated Sales Value (USD) using Price × Sales Volume to support sales-value analysis.
- Created an Engine Size Category column with the groups: Below 2L, 2L–2.99L, 3L–3.99L, and 4L+.
- Created a Price Segment column to classify vehicles into different price ranges: Under \$50K, \$50K–\$99K, \$100K–\$149K, and \$150K+.
- Renamed columns using business-friendly names where required for clearer dashboard visuals and DAX measures.
- Verified the transformed columns for errors and blank values before applying the changes using Close & Apply.







Data Modelling & Relationship :

DAX measures were created in Power BI to make the BMW sales data easier to understand and to display the most important business information in the dashboard. These measures help in analyzing sales performance, vehicle pricing, model performance, and other vehicle-related details.

The main DAX measures used in the dashboard are:

- Total Sales Volume – shows the total number of BMW vehicles sold.
- Average Sales Volume – shows the average sales volume across the available records.
- Average Price – gives the average price of the BMW vehicles.
- Total Models – shows the number of unique BMW models in the dataset.
- Total Records – shows the total number of sales records available for analysis.
- High Sales Count – identifies the number of records that fall under the high-sales category.
- Low Sales Count – identifies the number of records that fall under the low-sales category.
- Average Engine Size – shows the average engine capacity of the vehicles.
- Average Mileage – shows the average mileage recorded for the vehicles.
- Estimated Sales Value – represents the estimated value obtained by combining the vehicle price with its sales volume.

These measures are used throughout the dashboard to create KPI cards, sales trends, model comparisons, regional analysis, fuel-type comparisons, transmission analysis, price-segment analysis, and engine-size analysis. This makes it easier for users to interact with the dashboard and understand BMW sales performance from different perspectives.

DAX Measures and KPIs:

DAX Measure:

1. Maximum Sales Volume

Maximum Sales Volume =

MAX(bmw_sales_Data[Sales_Volume])

2. Total Sales Volume

Total Sales Volume =

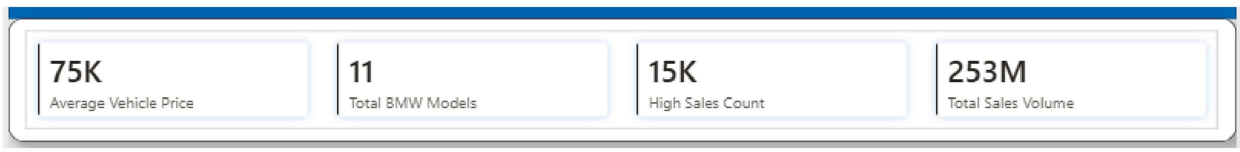
SUM(bmw_sales_Data[Sales_Volume])

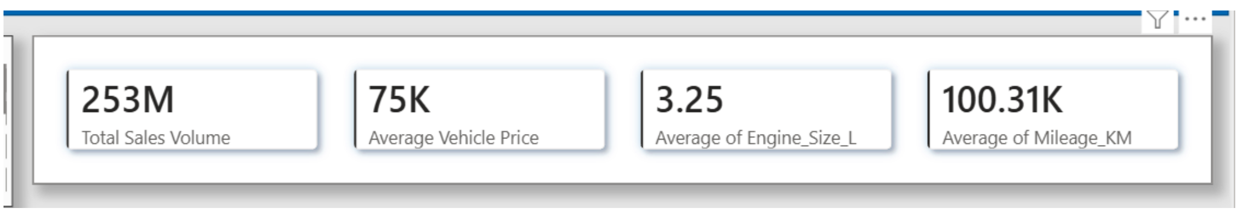
3. High Sales Percentage

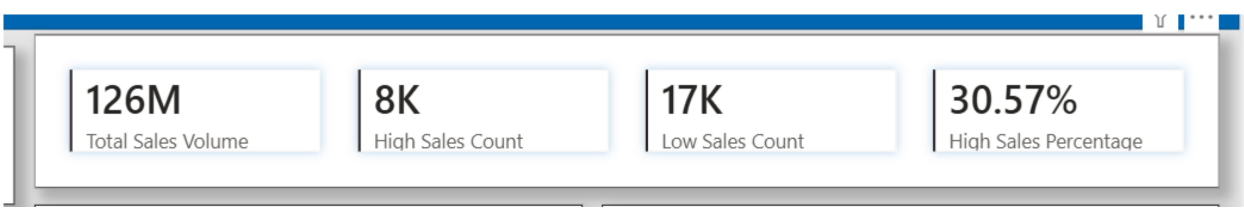
DIVIDE([High Sales Count],COUNTROWS(bmw_sales_Data),0)

In addition to the KPI measures, DAX was used to calculate maximum sales volume for yearly trend analysis and to calculate total sales volume and high-sales percentage for interactive analytical visuals. Other visualizations used Power BI's built-in aggregations such as Sum, Average, and Count.

KPI DAX & Its Screenshot Image :







1. Total Sales Volume

Total Sales Volume =SUM(bmw_sales_Data[Sales_Volume])

2. Average Vehicle Price

Average Vehicle Price =AVERAGE(bmw_sales_Data[Price_USD])

3. Total BMW Models

Total BMW Models =DISTINCTCOUNT(bmw_sales_Data[Model])

4. High Sales Count

High Sales Count
=CALCULATE(COUNTROWS(bmw_sales_Data),bmw_sales_Data[Sales_Classification] = "High")

5. Low Sales Count

Low Sales Count =
CALCULATE(COUNTROWS(bmw_sales_Data),bmw_sales_Data[Sales_Classification] = "Low")

6. High Sales Percentage

High Sales Percentage =DIVIDE([High Sales Count],COUNTROWS(bmw_sales_Data),0)

7. Average Engine Size

Average Engine Size =AVERAGE(bmw_sales_Data[Engine_Size_L]

8. Average Mileage

Average Mileage =AVERAGE(bmw_sales_Data[Mileage_KM])

9. Maximum Sales Volume

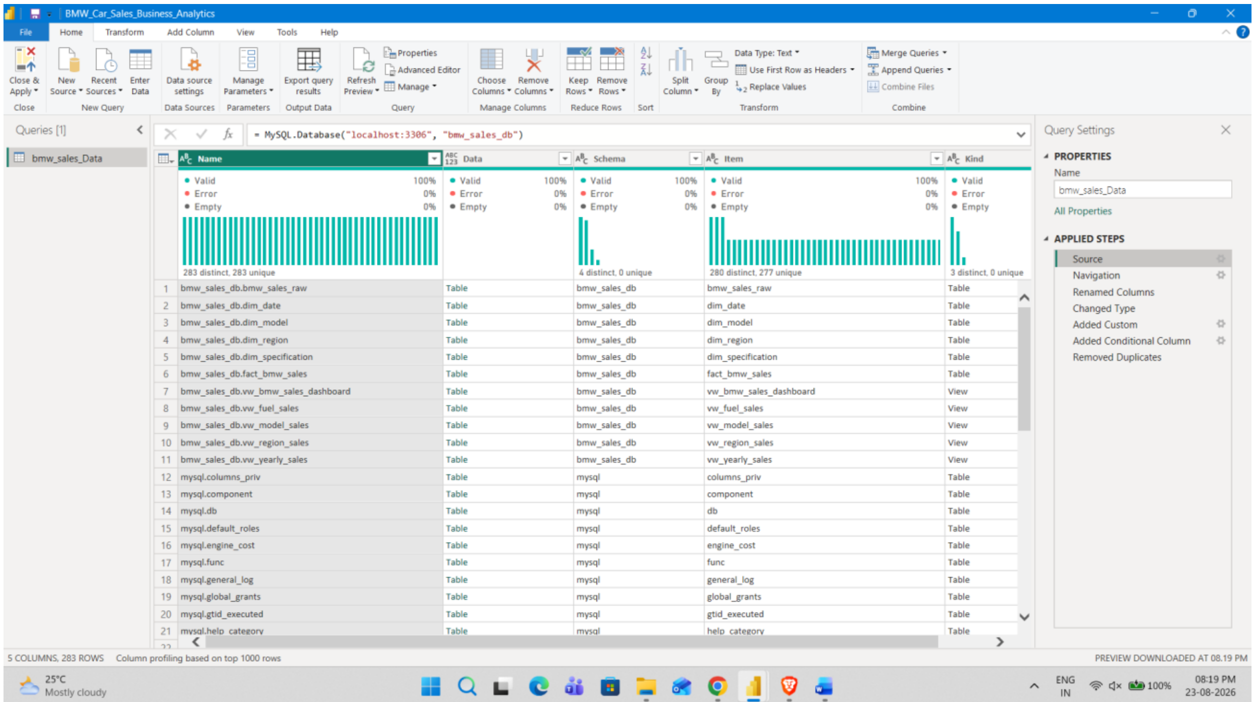
Maximum Sales Volume =MAX(bmw_sales_Data[Sales_Volume])

DAX measures were developed to calculate the key business metrics used across the three dashboard pages. These measures include total sales volume, average vehicle price, total BMW models, high and low sales counts, high-sales percentage, average engine size, average mileage, and maximum sales volume. The measures were used to create KPI cards, trend analysis, model comparisons, and regional and market performance visuals.

MySql Server Connection With Powerbi Desktop:

- Connected the **BMW sales database in MySQL** to Power BI and brought the required sales data into the report.
- Used the database connection to load the BMW sales information into Power Query for further analysis and dashboard development.
- Kept the connection refreshable so that the Power BI report can retrieve the latest available data from the MySQL database when refreshed.

Screenshot Image:

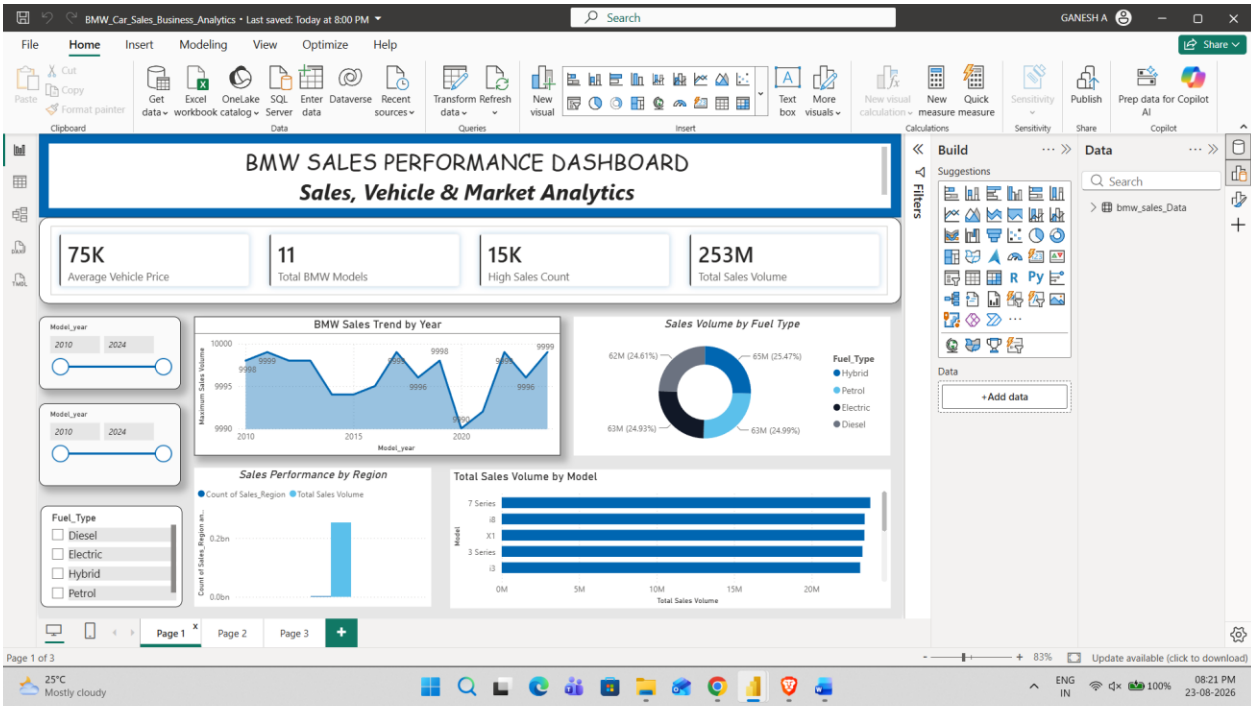


Interactive BMW Sales Dashboard

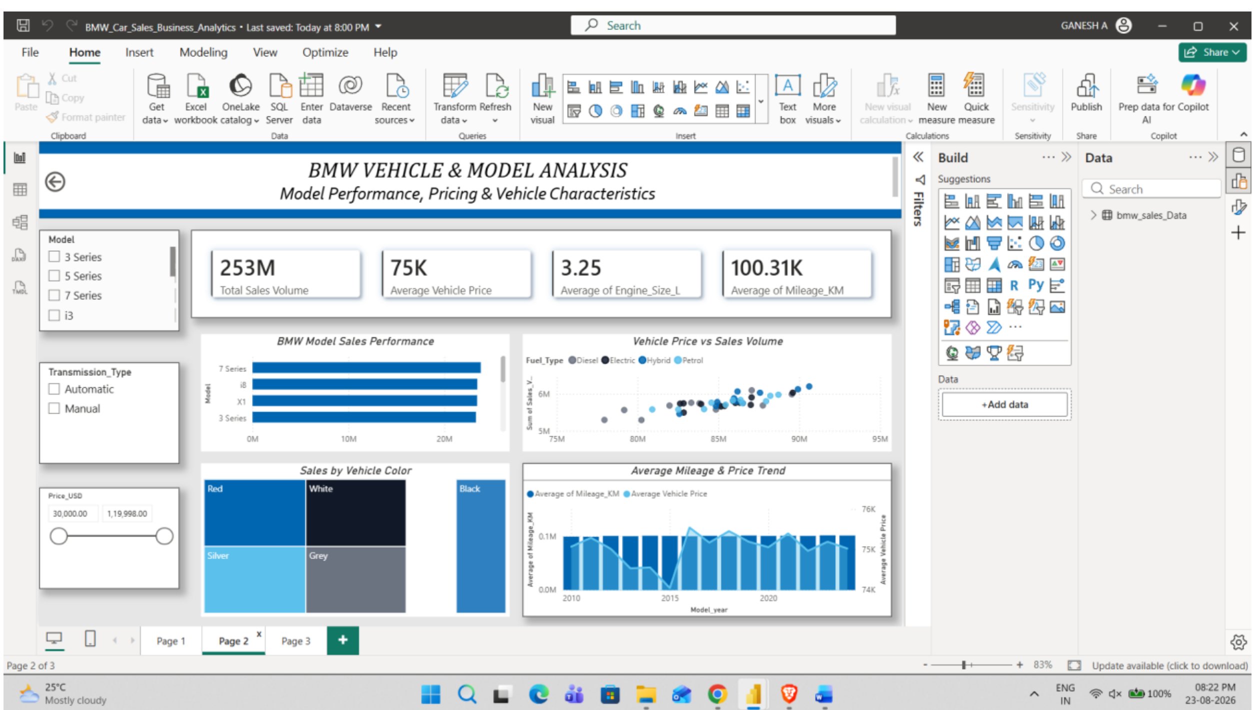
- The BMW Sales Analytics Dashboard gives an interactive view of overall sales performance, vehicle models, pricing, and regional market trends.
- Displays key metrics such as total sales volume, average vehicle price, total BMW models, and sales performance indicators.
- Used BMW Logo colour as it matches the dataset and company.
- Highlights the top-performing BMW models and compares their sales volumes.
- Shows year-wise sales trends to understand changes in BMW sales over time.
- Provides analysis of regional sales, fuel types, transmission types, vehicle colors, engine sizes, mileage, and price segments.
- Compares vehicle price with sales volume to identify useful sales patterns.
- Interactive slicers allow users to explore the data from different perspectives.
- The dashboard brings the main findings together in a simple visual format to support quick and informed business decisions.

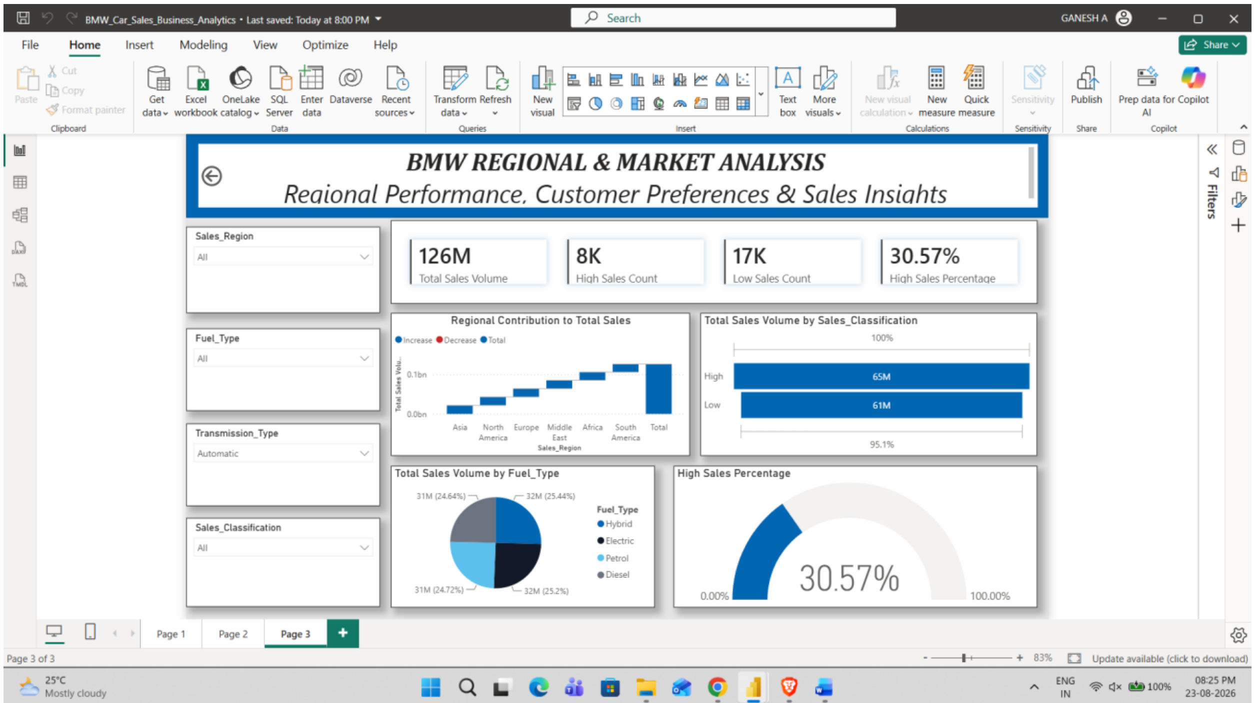
Screenshot Image:

Page 1 provides a quick overview of BMW sales performance using KPI cards, yearly sales trends, regional sales, fuel-type analysis, and top-performing BMW models. Interactive slicers allow users to filter the dashboard by model year, sales region, and fuel type.



Page 2 focuses on BMW model performance and vehicle characteristics, including top-selling models, price versus sales, engine size, mileage, vehicle color, transmission type, and price segments. Interactive slicers allow users to explore the analysis by BMW model, transmission type, and price segment.



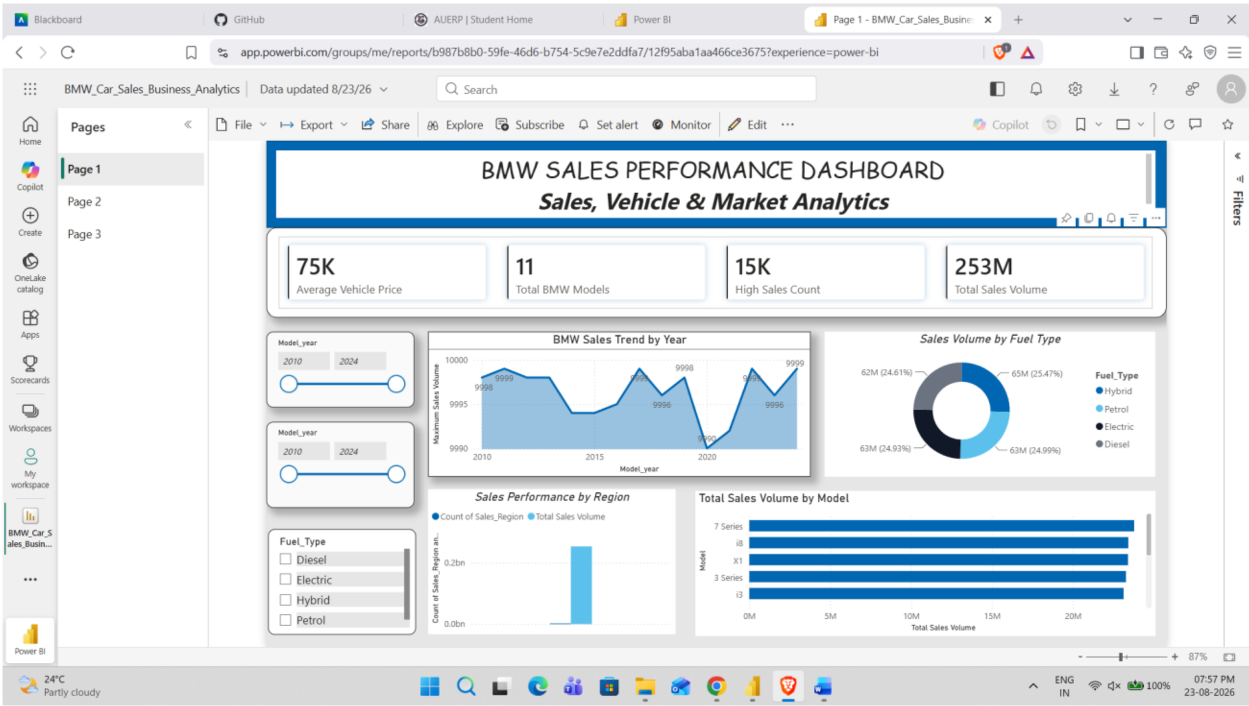


Page 3 focuses on regional sales performance and market preferences, covering sales by region, fuel type, transmission type, vehicle color, and sales classification. Interactive slicers allow users to filter the analysis by sales region, fuel type, transmission type, and sales classification.

Power BI Service Workspace:

- Published the Spotify Analytics Dashboard to the Power BI Service Workspace for online access and sharing.
- The dashboard is available in the workspace with interactive KPIs, charts, filters, and Spotify music analytics.
- The published report can be accessed through the following Power BI Service link: View BMW Sales Dashboard in Power BI Service
- The Power BI Service enables the dashboard to be viewed, shared, and accessed online from the workspace.

Screenshot Image:



<u>Business Question Answered:</u>

I. What is the overall BMW sales performance?

Answer: The dashboard shows a total sales volume of approximately 253 million across the dataset. There are 11 BMW models, while the average vehicle price is around $75K.

II. Which BMW models have the highest sales volume?

Answer: The 7 Series, i8, X1, 3 Series, and i3 appear among the leading models in the Top Model analysis. The bars are very close in size, indicating that sales are relatively competitive across the leading models.

III. How does BMW sales volume change across model years?

Answer: The yearly trend is fairly stable, staying close to 10,000 units per year in the displayed maximum-sales analysis. There are some fluctuations, with a noticeable dip around 2020 followed by recovery in the following years.

IV. Which fuel type contributes the most to sales?

Answer: The four fuel types have a very similar contribution. Hybrid is slightly ahead at about 65M (25.47%), followed by Petrol at about 63M (24.99%), Electric at about 63M (24.93%), and Diesel at about 62M (24.61%). This indicates that no single fuel type dominates the dataset.

V. Is there a relationship between vehicle price and sales volume?

Answer: The scatter plot shows a generally positive pattern between vehicle price and sales volume in the displayed data, although the relationship is not perfectly uniform. This suggests that higherpriced vehicles can still achieve strong sales volumes, so price alone does not appear to determine sales performance.

VI. Which engine and vehicle characteristics describe the dataset?

Answer: The average engine size is approximately 3.25 L, while the average mileage is around 100.31K KM. These values provide a general picture of the vehicle characteristics represented in the dataset.

VII. Which vehicle colors have strong sales representation?

Answer: The vehicle-color treemap shows that Red, White, Black, Silver, and Grey have substantial sales representation. Red and White occupy some of the larger sections, indicating strong representation among the recorded sales.

VIII. Which regions contribute strongly to overall sales?

Answer: The regional contribution chart shows that South America has the largest contribution, followed by Africa, Middle East, Europe, North America, and Asia in the displayed cumulative analysis.

IX. What is the difference between high-sales and low-sales performance?

Answer: The dashboard shows approximately 65M sales volume for the High classification and around 61M for the Low classification. The Page 3 KPI also shows a 30.57% High Sales Percentage, indicating that high-sales records represent roughly three-tenths of the records under the current dashboard context.

X. What happens when the dashboard filters are applied?

Answer: The KPI values change based on the selected model year, region, fuel type, transmission type, model, price range, and sales classification. This allows users to isolate particular parts of the BMW sales data and compare performance under different business conditions.

Business Insights and Recommendation:

Key Insights

- The analysis shows that BMW sales are spread across 11 models, with the leading models such as the 7 Series, i8, X1, 3 Series, and i3 making up a significant part of the recorded sales.
- Total sales volume is approximately 253 million, while the average vehicle price is around $75K, showing that BMW operates across a wide range of vehicle values.
- The yearly analysis shows that sales volume remains fairly stable across the model years, although some years show noticeable increases and decreases.
- The four fuel types have a very balanced sales contribution. Hybrid leads slightly at about 65M (25.47%), while Petrol, Electric, and Diesel remain close behind.
- The price-versus-sales analysis indicates a generally positive relationship between vehicle price and sales volume, although price alone does not fully explain sales performance.

- The average vehicle has an engine size of approximately 3.25 L and mileage of around 100.31K KM, providing a useful overview of the vehicles represented in the dataset.
- Red, White, Black, Silver, and Grey show strong representation in the sales data, with Red and White among the larger categories.
- South America has the largest regional contribution in the displayed analysis, followed by Africa, the Middle East, Europe, North America, and Asia.
- High-sales records account for approximately 30.57% according to the dashboard KPI, while the overall sales volume of the High category is around 65M, compared with roughly 61M for the Low category.
- The dashboard demonstrates that sales performance can vary when users apply filters for model year, region, fuel type, transmission, model, price range, and sales classification, making the analysis more useful for different business scenarios.

Recommendations

- Focus marketing and inventory planning on the BMW models and regions that consistently show stronger sales performance, while reviewing weaker segments for potential improvement.
- Since the fuel types have relatively similar sales contributions, BMW can maintain a balanced approach to different fuel categories while monitoring how customer preferences change over time.
- Use the relationship between vehicle price and sales volume to support pricing decisions and identify price ranges that maintain strong demand.
- Give additional attention to the strongest-performing regions, especially South America, by studying their product preferences and adapting regional sales strategies accordingly.
- Use the sales classification results to identify high-performing vehicle segments and develop targeted promotional or inventory strategies around them.
- Continue monitoring vehicle characteristics such as engine size, mileage, color, and transmission to better understand which combinations are associated with stronger sales.
- Use the dashboard's interactive filters regularly to compare different market segments before making decisions about pricing, marketing, inventory, and product focus.

Conclusion

This project developed an end-to-end BMW Sales Business Intelligence solution using MySQL, Power Query, DAX, and Power BI. The BMW sales data was connected to MySQL, analysed using SQL, transformed in Power Query, and then presented through an interactive three-page Power BI dashboard. The dashboard brings together sales volume, pricing, vehicle characteristics, model performance, and regional market information in one place. The final dashboard helps users move from a high-level sales overview to detailed vehicle and regional analysis through interactive KPIs, slicers, and visualizations. The results provide useful support for understanding BMW sales patterns and can help management make more informed decisions related to product planning, pricing, marketing, and regional strategy.